Mojtaba Kalantari

RF & Microwave Engineer

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PROFILE

RF and microwave engineer working on passive RF components, antenna design, and electromagnetic simulation. Experience spans academic research and industrial development, including prototype manufacturing, RF measurements, and hardware validation.

EDUCATION

M.Sc. in Electrical Engineering - Telecommunications, Fields and Waves

K. N. Toosi University of Technology | Sep 2023 - Feb 2026

Thesis: Design and Simulation of a Diplexer in Two Frequency Bands (1710-1785 MHz and 1805-1880 MHz) Using the Coupling Matrix Method. Thesis grade: 20/20.

B.Sc. in Electrical Engineering - Telecommunications

K. N. Toosi University of Technology | Sep 2019 - Sep 2023

Thesis: Design and Construction of a Cross-Dipole Antenna with Reduced Interaction Effects on Nearby Antennas for Base Station Applications. Thesis grade: 20/20.

ENGINEERING EXPERIENCE

Research & Development

WITEM Laboratory | Mar 2023 - Present

Coupling-matrix synthesis and mechanical diplexer design; base-station antenna elements, true-time-delay phase shifters, and fixed-beam arrays. Work includes electromagnetic design, manufacturing, and testing.

RF Hardware & Antenna Projects

Honar Mouj Nasir (H.M.N.) | Apr 2023 - Present

Cellular repeater development, indoor and outdoor antennas, multilayer RF PCBs (1-4 layers), prototype assembly, RF measurements, troubleshooting, bills of materials, and manufacturing support. Additional work on absorbers and microwave reflector links.

R&D Engineer

Payaco Communication Industries | Mar 2022 - Nov 2022

RF troubleshooting, performance improvement, and optimization of testing and assembly processes. Delivered a six-session technical course on BTS links, PIM, matching, and radiation characteristics.

Antenna Engineering Intern

MOJPA | Jun 2021 - Sep 2021

Design, manufacturing, and testing of an X-band circularly polarized horn antenna with a turnstile OMT feed.

SELECTED ENGINEERING PROJECTS

Coupling-Matrix Diplexer

Microwave filters | M.Sc. thesis

Design and EM simulation of a coaxial-resonator diplexer for 1710-1785 MHz and 1805-1880 MHz, connecting network synthesis to a practical resonator geometry.

Crossed-Dipole Antennas & Antenna Interaction

Antenna design, manufacturing, and testing

Crossed-dipole elements for 690-960 MHz, 1690-2690 MHz, and 2300-3800 MHz. Work on parasitic elements, reduced interaction with nearby antennas, and co- and cross-polarization tuning.

True-Time-Delay Phase Shifter & Fixed-Beam Array

Feed networks and arrays

Nine-element true-time-delay phase shifter for 690-960 MHz with 12-degree electrical tilt. Twelve-element fixed-beam array for 2300-3800 MHz with 6-degree elevation tilt.

Microwave Reflector Links & Absorbers

Microwave antenna development

Hat-fed reflector antennas at 15, 18, 23, and 26 GHz using 2-, 3-, and 4-foot reflectors. Absorber design, manufacture, and testing for backlobe reduction; pyramidal absorber work for 1-12 GHz.

RF Components & Other Antennas

Passive components and specialist antennas

Microstrip diplexer work for 1690-2200 MHz and 2400-2690 MHz; K-, Ku-, and Ka-band microwave filters; and a 406 MHz omnidirectional antenna for weather-balloon communication.

PUBLICATIONS

An improved printed Palm crossed dipole antenna with reduced interference on adjacent elements. *Manuscript under review, 2026.*

TEACHING & TECHNICAL TRAINING

Teaching assistant experience: Microwave Engineering (2026); Electromagnetic Fields and Waves (2025-2026); Antenna I & II (2023-2025); Signals and Systems (2023); Digital Logic (2022-2023). Activities included problem-solving sessions, assessment, electromagnetic analysis, simulation training, and project support.

TECHNICAL SKILLS

Software: CST Studio Suite, Ansys HFSS, Keysight ADS, MATLAB, Altium Designer, SolidWorks, Python, and FEKO.

Engineering: Antenna and array design, microwave filter synthesis, RF measurements, mechanical design, RF PCB design, prototype assembly, and hardware validation.

LANGUAGES

Persian (native); English; German.